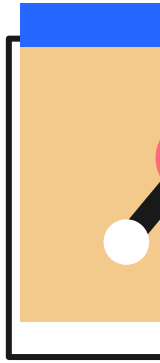
Find Close-By Events: Spot Drop



Ezequiel, Shenrui, Siddhartha

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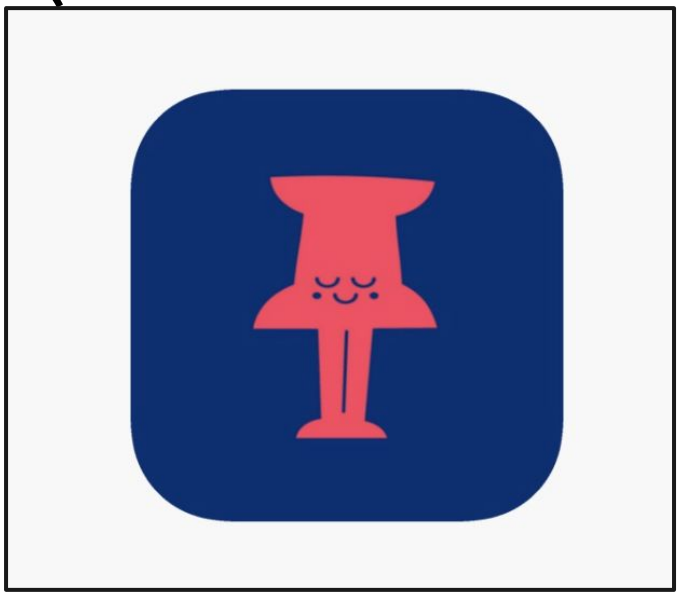
06

Where Are We? & Demo

What is Spot Drop?

SpotDrop is a transformative platform (a mobile application) designed to solve the universal business challenge of vacant spots or seats across virtually every service and experience sector.

Businesses often lose money when their slots, seats, or appointment times go unfilled, and SpotDrop aims to turn these empty spaces into opportunities by creating a dynamic marketplace.



Why For Businesses?

- Identify Empty Spots: Businesses can identify available slots hours or minutes before a class or event, or immediately after a cancellation.

- Post a "Drop": They can then enter the available spots and set a discount.

- Reach Local Users: Nearby users are instantly notified via push notification.

- Receive Bookings & Payment: Payments are processed securely and directly through the app.

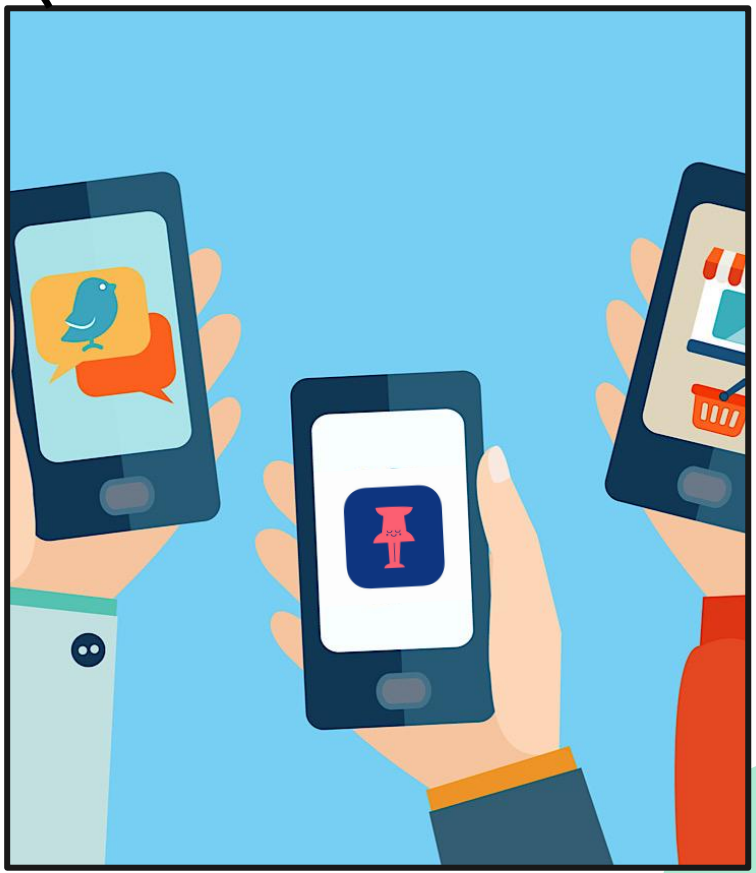


Why For Users?

• Get Notified: Users receive real-time alerts for discounted classes, appointments, and events in their vicinity.

• Browse Deals: They can view available options and discounts.

• Book & Pay: Users can securely book and pay through the app.



What Are We Doing?

Partnership & Integration

Connecting with third-party websites to make it easier for users to know what events are happening.

Customer Experience Optimization:

Improving booking, and search journeys to deliver a smoother, more intuitive user experience.

Data & Intelligence

Developing AI-driven tools to automatically gather and organize events' data with internal dashboards for quality control and oversight.

Timeline

1. Free Event Website Scraper BaseLine (SpotHub) — Oct 18
2. Adding Image Crawling to the Scraper — Oct 24
3. Add Latitude and Longitude Information Extraction to the Scraper — Nov 7
4. Adding Map Pins for Events — Nov 16
5. Creating categories/interests & UI/UX changes to the profile page — Nov 18
6. Adding Filter for Events (Map and Event Page) Based on user schedule — Nov 20
7. Adding AI-Categorizations & Database Persist to the Scraper — Nov 26
8. Prototype Chat-Based Assistant For Events — Dec 5

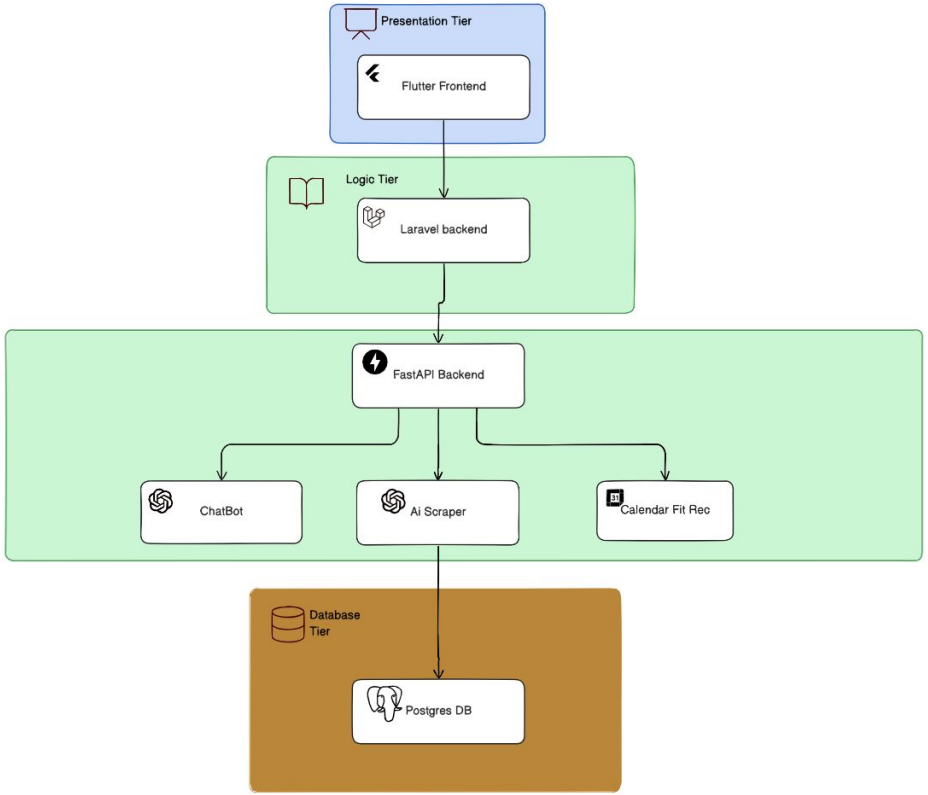
Tech Stack

Our tech stack:

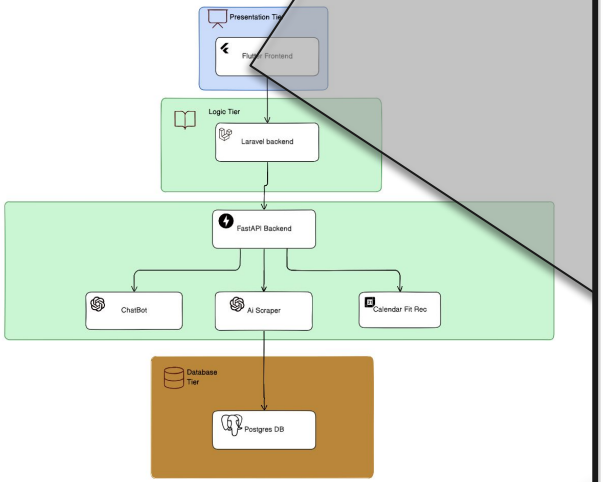
- Flutter
- Laravel/PHP
- Python
- PostGresQL
- AWS ECS Lambda
- OpenAI API

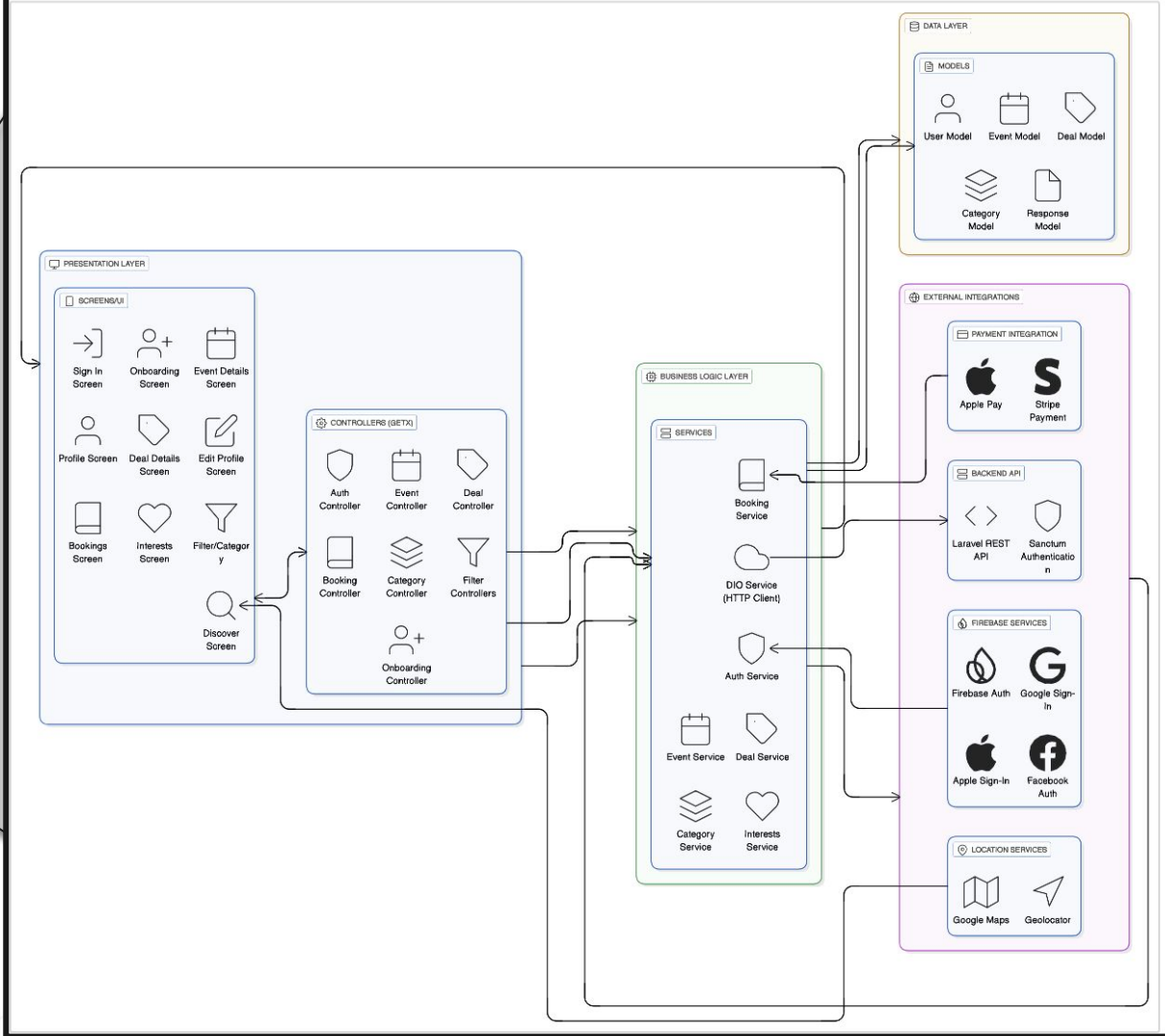


Architecture Diagrams - high level overview

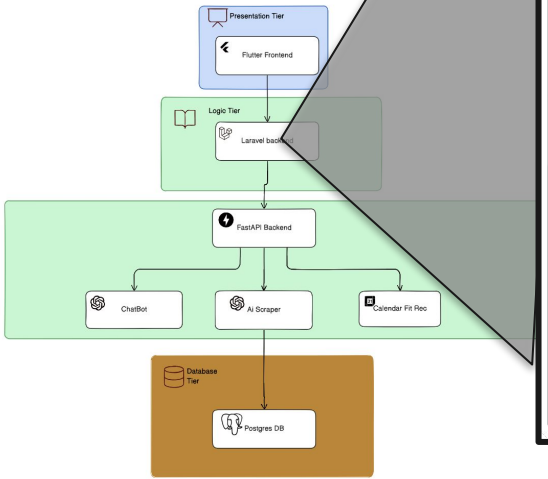


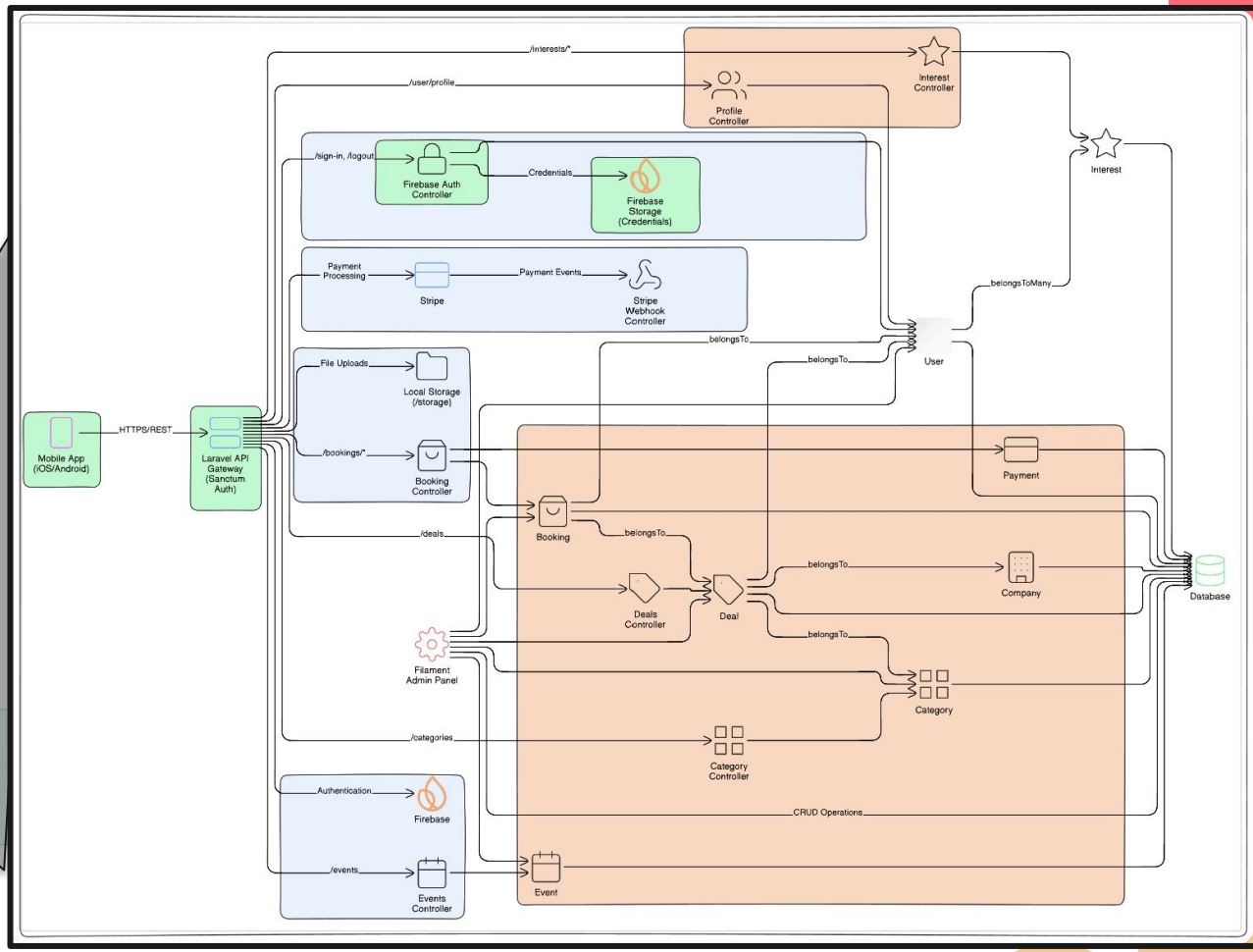
Flutter Frontend Diagram



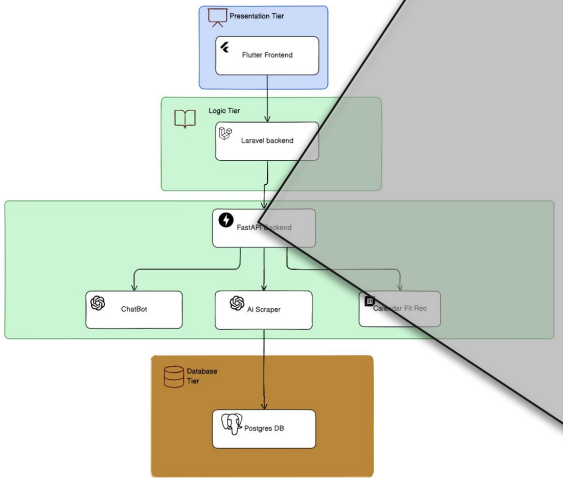


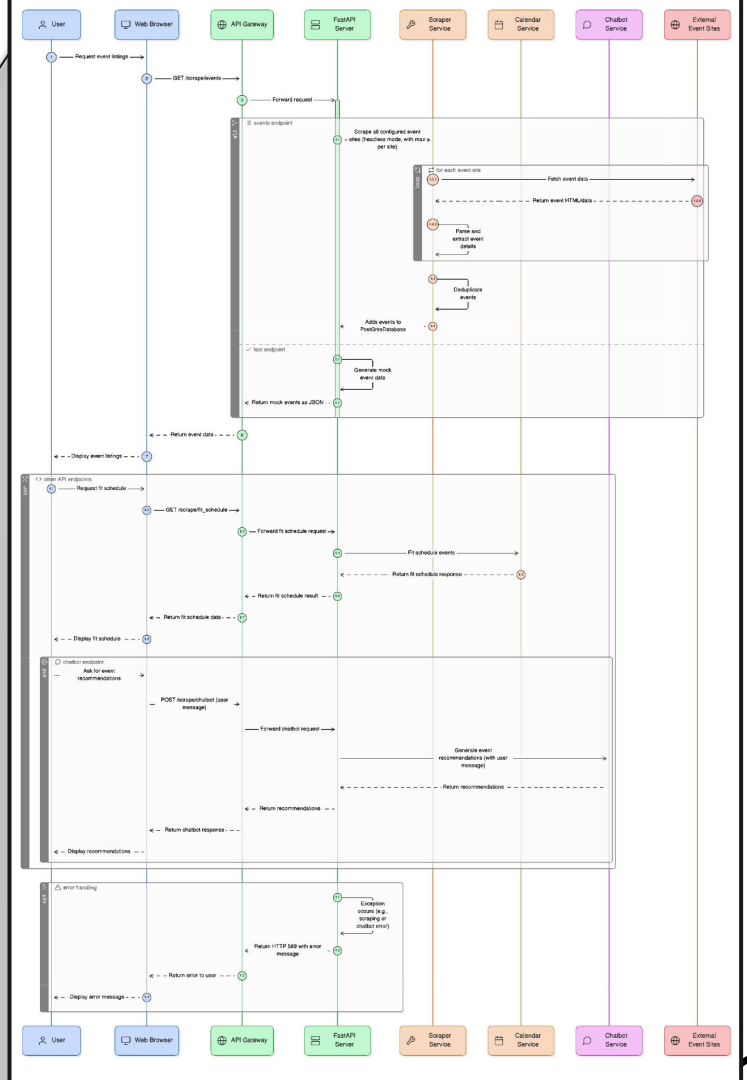
Laravel Backend Diagram



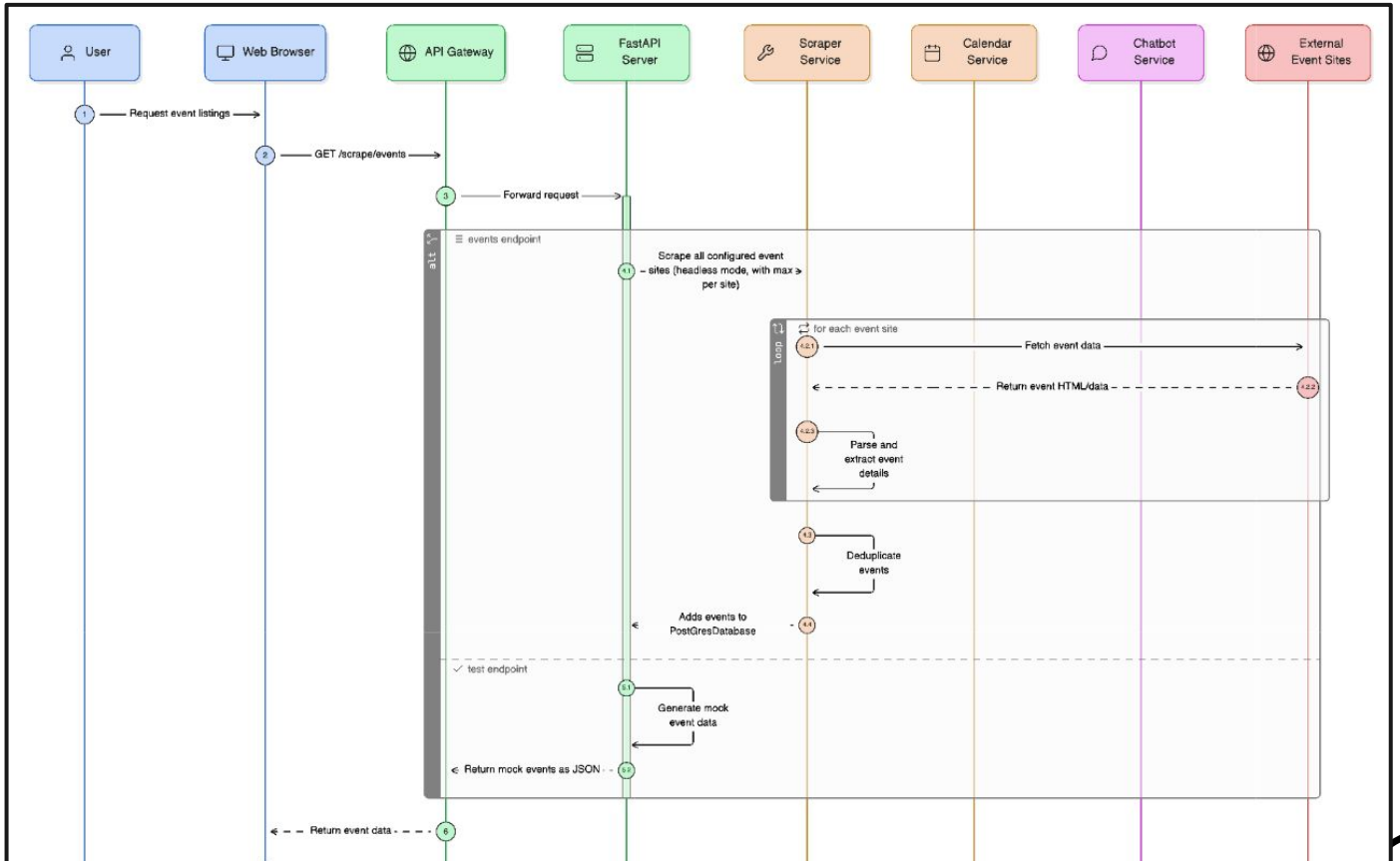


FastAPI Diagram

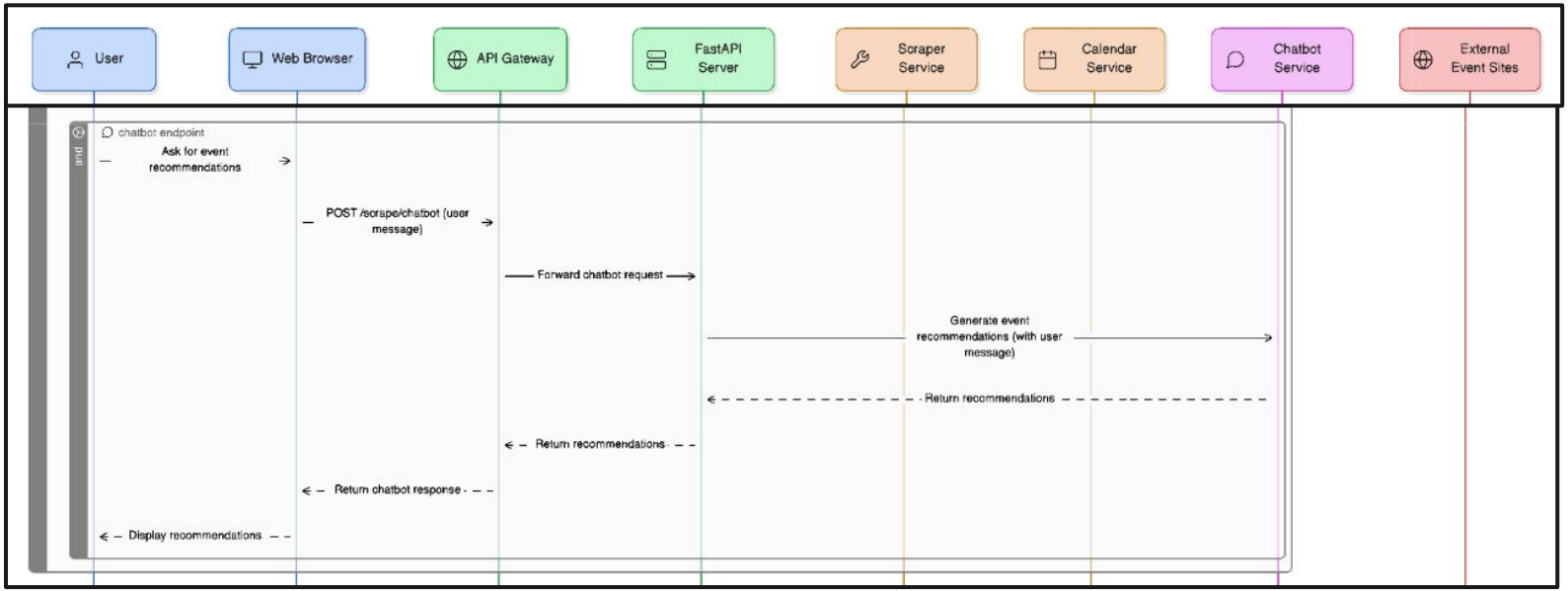




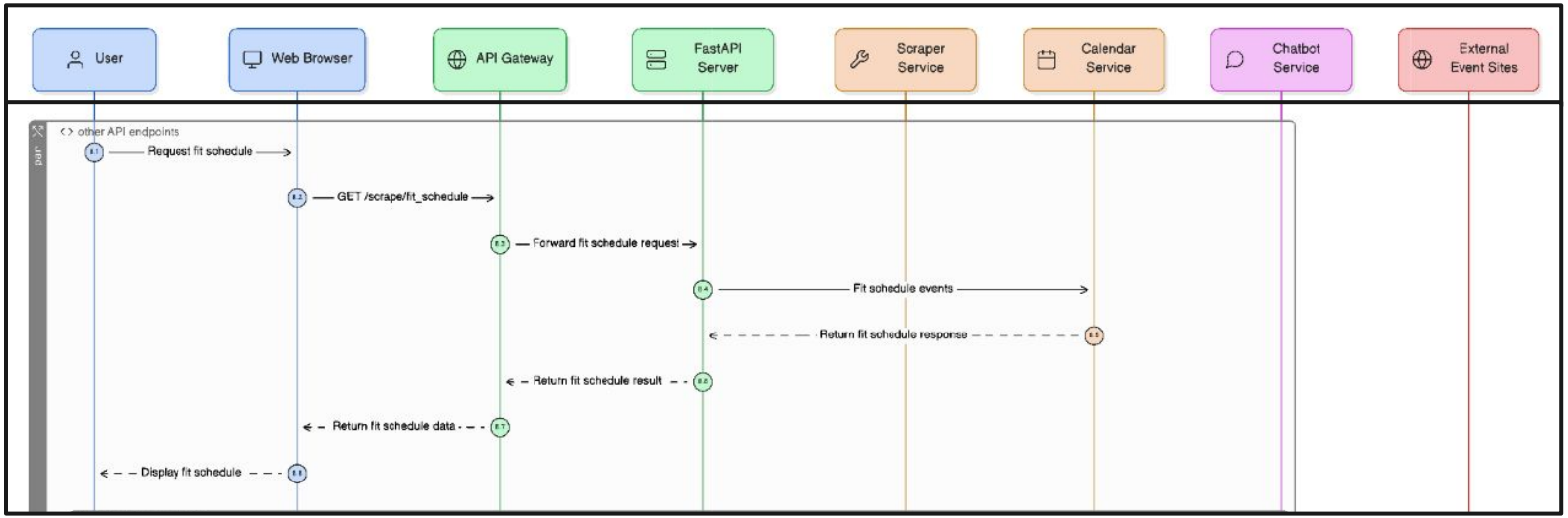
AI-Scraper Endpoint



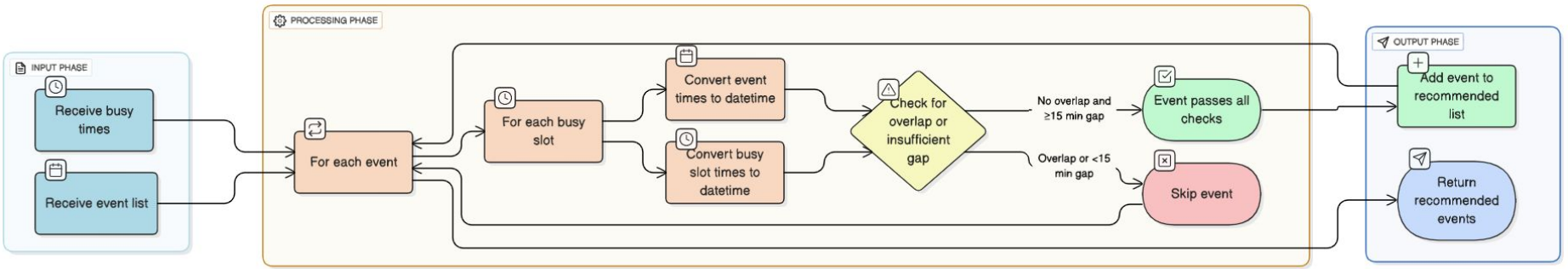
Chatbot Endpoint



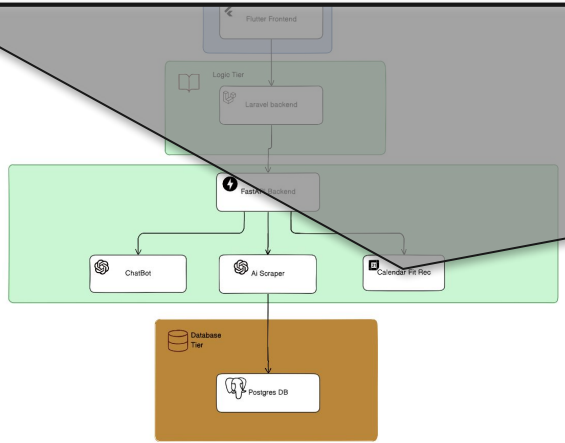
Calendar_rec_events Endpoint



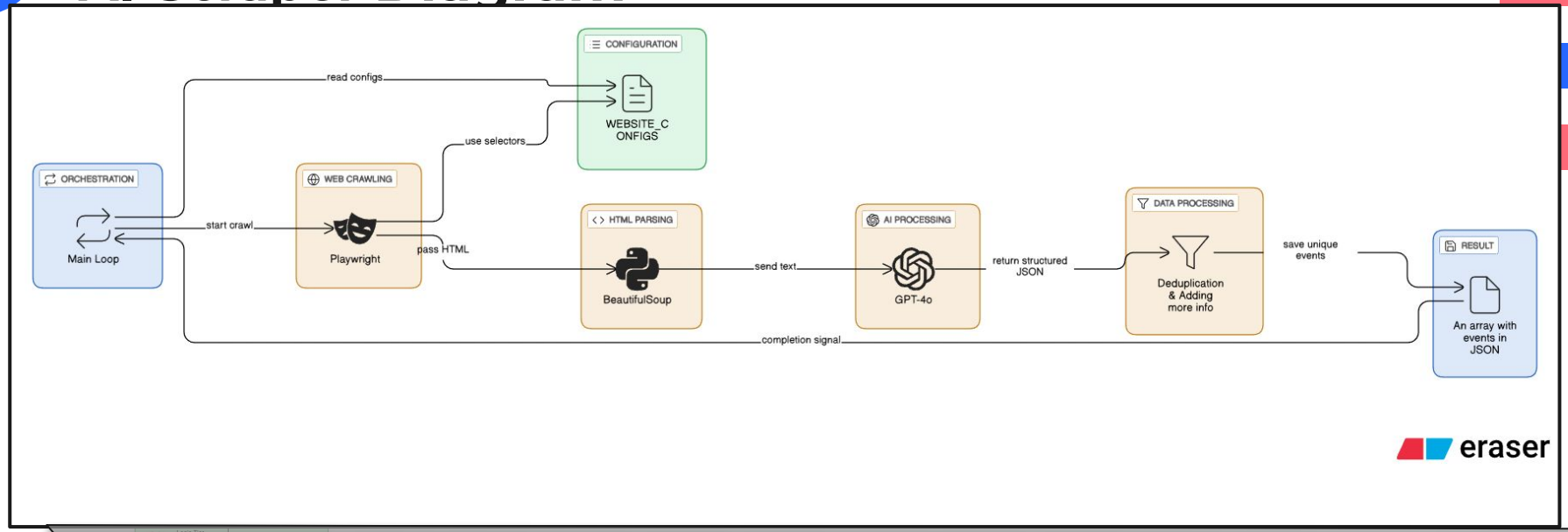
Calendar_rec_events

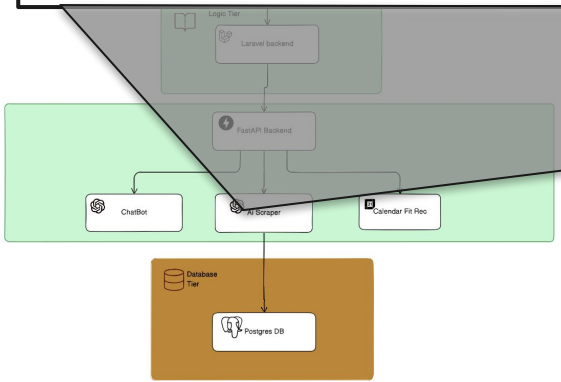


eraser

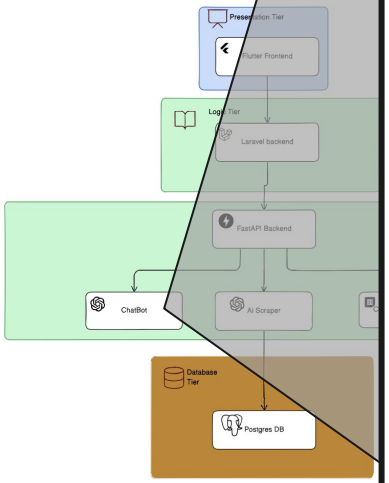


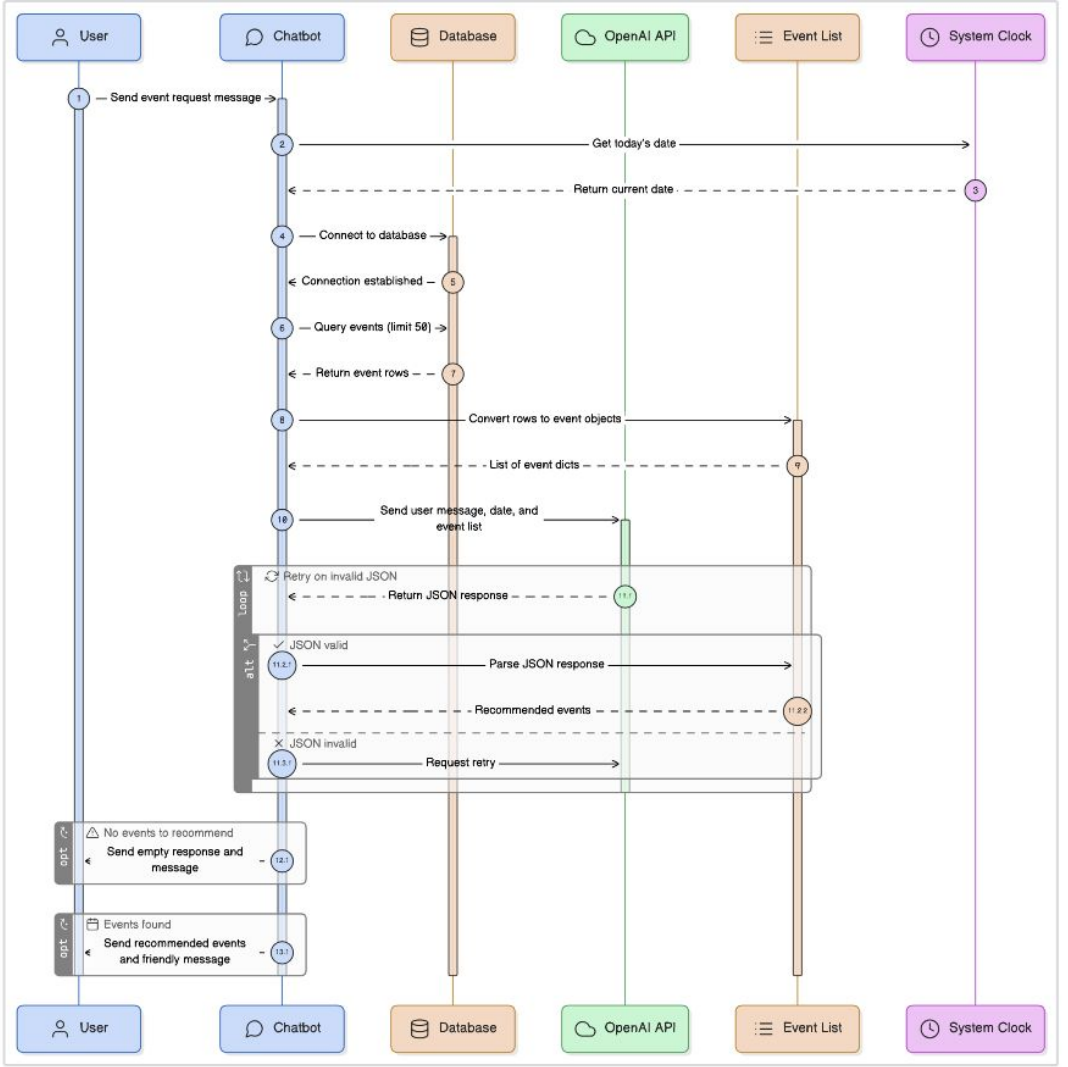
AI-Scraper Diagram





ChatBot Diagram



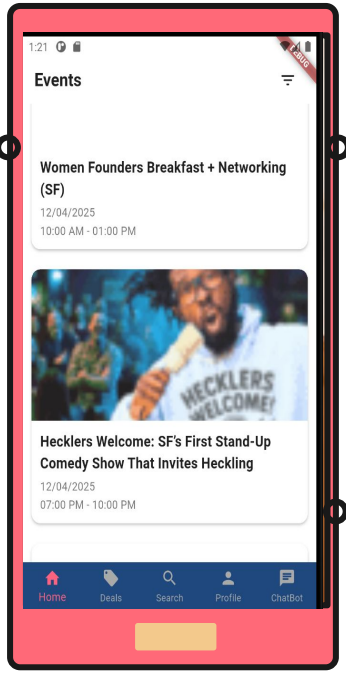


Where Are We?
& Demo

Where Are We? & Demo

Home

This is the main home page where a user can see events happening in their area today



Features

On this page you can press on events to get more information and can visit the event sites

Updates

Currently some images are not formatted properly and come as data:srv. We still need to find a way to fix this

Where Are We? & Demo

Info when event pressed



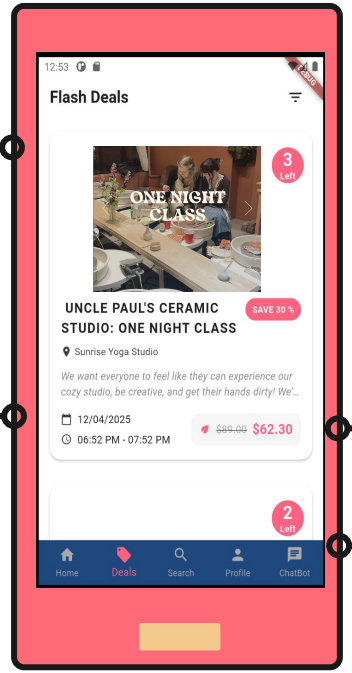
Where Are We? & Demo

Flash Deals

The Flash Deals will be showed here in the deals page

Time Information

When is the deal expected to be start and end



Price

Share the price with users

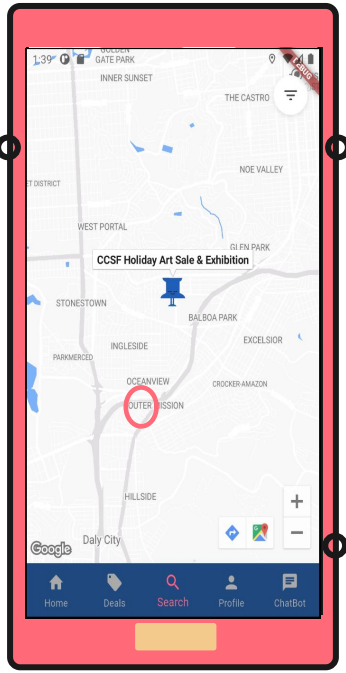
Updates

Right now we don't have any businesses partnering with. This is just a mock deal

Where Are We? & Demo

Events and deals pin

Events and deals show up on the map



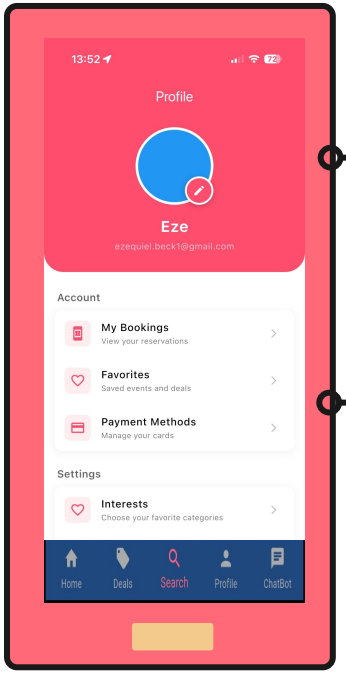
Features

When you press a pin it gives the title of the event

Updates

In the future you could press the pin and it would give you the specific details

Where Are We? & Demo



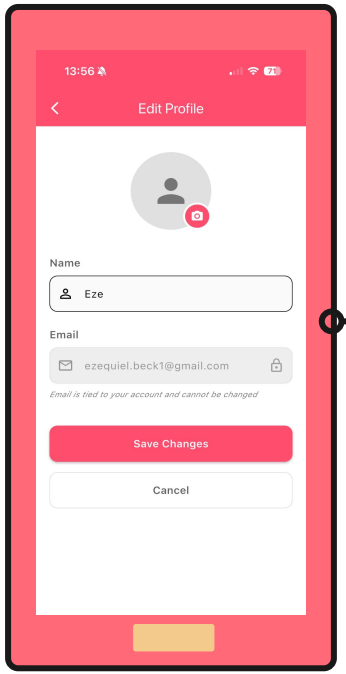
Edit Profile

Edit Profile Button for User Experience

Account Buttons

Buttons to access account settings like bookings, favorites, payment methods

Where Are We? & Demo



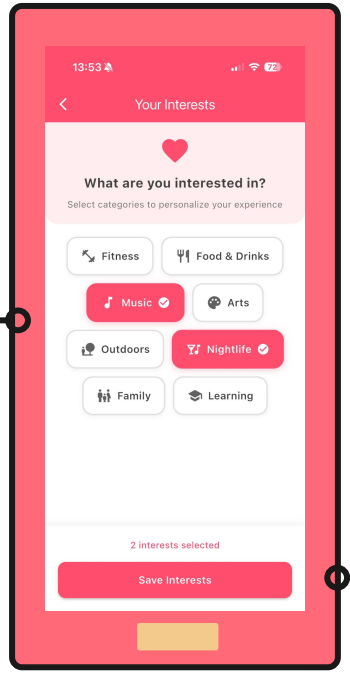
Name Edit

For now, you can only edit your name.

Where Are We? & Demo

Choose your favorite interests

Choose your favorite interests for deals/events



Save button

Save your interests. You can come back and change them later.

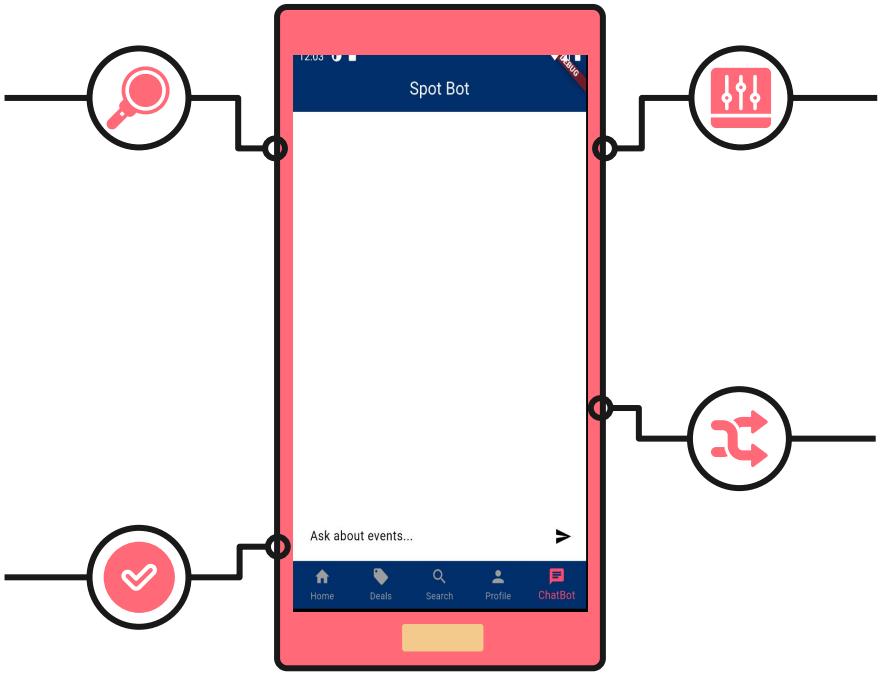
Where Are We? & Demo

SpotBot

Simple chat bot that you can ask questions about events

Availability

Currently this is still a work in progress and not live on the app



Features

It has current date and all events from the database so you can ask about events

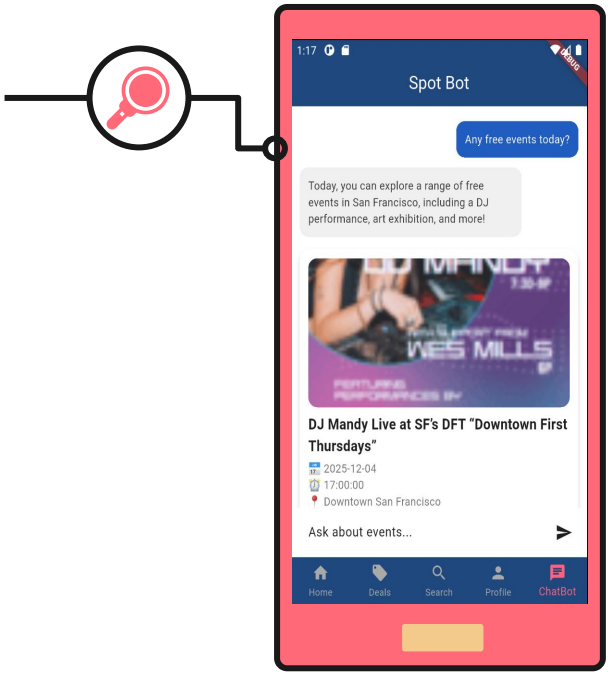
Updates

Since there are not many events we are just sending the events to the LLM , also we will add category preference recommendations later

Where Are We? & Demo

Example question

Gives a response and the info about events



Challenges

One big challenge we had was the learning curve at the beginning when we started to work on the project. PHP is definitely a language that took us some time to learn in order to get our backend working and have our features up and running. Additionally, Flutter was also a bit complicated in learning since it has a different style in comparison to other languages that are used to build mobile app.

Another challenge we faced throughout the semester was token usage with the OpenAI API. At first, we didn't realize and didn't really look into how many tokens the API was using per call for our features. We first realized this after exhausting our token limit. As a result, we added a test limit for the API call.

Due to the senior capstone project being our first real experience on developing multiple features at the same time for one product, one challenge we had was with source control. Sometimes we would forget to pull and create merge conflicts which were a real headache to solve and correct in order to continue development on the app.

Future Implementations

Social Layer / Meetups

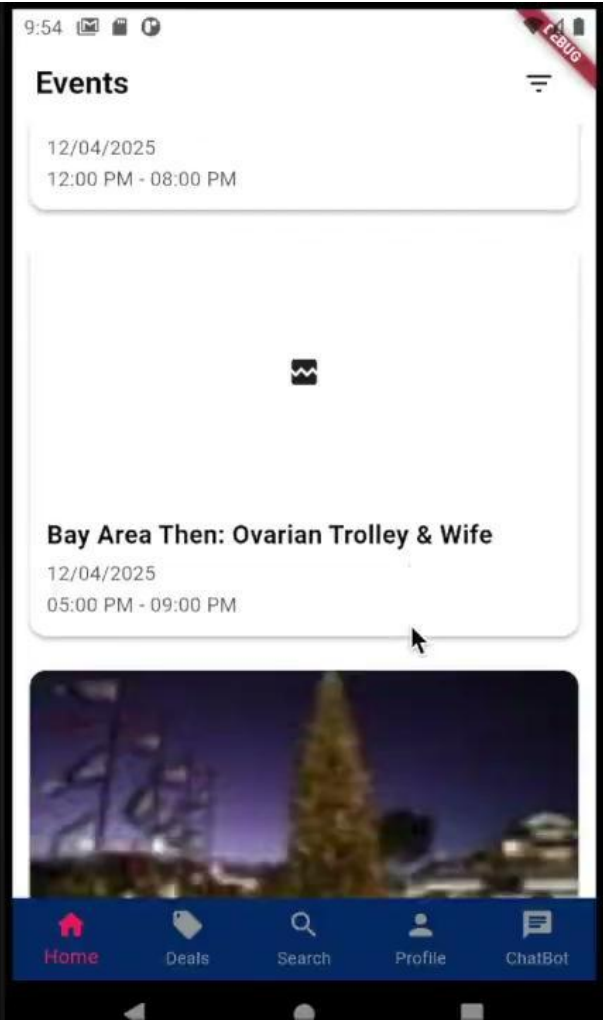
- **Personality & vibe matching**: AI matches attendees for potential meetups based on profiles, text sentiment, or shared categories.
- **Safety filtering**: Flag fake accounts or bots trying to exploit meetup features.
- **Conversation starters**: Generate icebreaker suggestions between like-minded users.
- **Deals implementation**: Currently have a few businesses on board for the deals, still have to make a way to get last-minute business last minute deals and display them on the app.

User Engagement & Retention

- **Behavior prediction**: AI models detect churn risk (user hasn't opened the app in X days, no saved events) and send personalized nudges.
- **Dynamic feed ranking**: Events aren't just chronological → ranked by relevance, proximity, popularity, and user intent.
- **Smart onboarding**: AI asks minimal questions at signup, then fills in gaps by inferring preferences from behavior.

DEMO

!!!


9:54
DEBUG
Events
12/04/2025
12:00 PM - 08:00 PM
Bay Area Then: Ovarian Trolley & Wife
12/04/2025
05:00 PM - 09:00 PM
Home
Deals
Search
Profile
ChatBot

Thanks!